ECE 2305 – Introduction to C Programming
Programming Tutorial 25
Pointers and 2D Arrays

Execute the following statements:

```
double a[2][2] = {{2.3, 4.1}, {6.2, 8.3}};
double b = 258.6;
double* c;
double* d;
c = a[0];
d = &b;
```

and fill in the following memory map.

Variable Name		Address	Value
a[0][0]			
a[0][1]			
a[1][0]			
a[1][1]			
b			
*c			
*d			

Next, execute the following code and fill in the memory map again. Take note of the changes.

```
c = c + 3;
*c = 10.4;
*d = 517.2;
```

Variable Name	Address	Value
a[0][0]		
a[0][1]		
a[1][0]		
a[1][1]		
b		
*c		
*d		

Input

```
//ECE 2305 - Tutorial 25
#include <iostream>
#include <iomanip>

using namespace std;
int main() {
    double a[2][2] = { {2.3, 4.1}, {6.2, 8.3} };
    double b = 258.6;
    double* c;
    double* d;

    c = a[0];
    d = &b;

    cout << left << setw(12) << "Variable" << setw(20) << "Address" << "Value" <<
endl;
    cout << endl;
    cout << "a[0][0]" << setw(5) << "" << setw(20) << &a[0][0] << a[0][0] <<
endl;
    cout << "a[0][1]" << setw(5) << "" << setw(20) << &a[0][1] << a[0][1] <<
endl;
    cout << "a[1][0]" << setw(5) << "" << setw(20) << &a[1][0] << a[1][0] <<
endl;
    cout << "a[1][1]" << setw(5) << "" << setw(20) << &a[1][1] << a[1][1] <<
endl;
    cout << endl;
    cout << setw(12) << "b" << setw(20) << &b << b << endl;
    cout << setw(12) << "*c" << setw(20) << c << *c << endl;
    cout << setw(12) << "*d" << setw(20) << d << *d << endl;
    cout << endl;
```

```
    cout << "After pointer arithmetic and assignments" << endl;
    cout << "---------------------------------------\n";

    c = c + 3;
    *c = 10.4;
    *d = 517.2;

    cout << left << setw(12) << "Variable" << setw(20) << "Address" << "Value" <<
endl;
    cout << endl;
    cout << "a[0][0]" << setw(5) << "" << setw(20) << &a[0][0] << a[0][0] <<
endl;
    cout << "a[0][1]" << setw(5) << "" << setw(20) << &a[0][1] << a[0][1] <<
endl;
    cout << "a[1][0]" << setw(5) << "" << setw(20) << &a[1][0] << a[1][0] <<
endl;
    cout << "a[1][1]" << setw(5) << "" << setw(20) << &a[1][1] << a[1][1] <<
endl;
    cout << endl;
    cout << setw(12) << "b" << setw(20) << &b << b << endl;
    cout << setw(12) << "*c" << setw(20) << c << *c << endl;
    cout << setw(12) << "*d" << setw(20) << d << *d << endl;

    system("pause");
    return 0;
}
```

Output

```
C:\Users\anmal\Desktop\ECE
Variable    Address             Value

a[0][0]     000000EED1F8F008    2.3
a[0][1]     000000EED1F8F010    4.1
a[1][0]     000000EED1F8F018    6.2
a[1][1]     000000EED1F8F020    8.3

b           000000EED1F8F048    258.6
*c          000000EED1F8F008    2.3
*d          000000EED1F8F048    258.6

After pointer arithmetic and assignments
---------------------------------------
Variable    Address             Value

a[0][0]     000000EED1F8F008    2.3
a[0][1]     000000EED1F8F010    4.1
a[1][0]     000000EED1F8F018    6.2
a[1][1]     000000EED1F8F020    10.4

b           000000EED1F8F048    517.2
*c          000000EED1F8F020    10.4
*d          000000EED1F8F048    517.2
Press any key to continue . . .
```